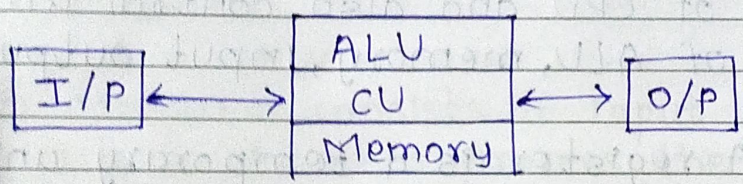


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Unit 1: Introduction

- A computer is a combination of hardware and software resources which integrate together and provide various functionalities to user. Hardware are physical component of computer. Processor, memory, monitor, keyboard etc. While software is set of program or instruction that are required by hardware resources to function properly.
- Digital computer - It can be define as programmable device which read binary data pass as instruction, process this binary data and display a calculated digital output.

* Functional component of computer -



- 1) Input unit consists of input devices that are attached to the computer that devices take input and convert it into binary for CPU. Some of input devices are keyboard, mouse, joystick, scanner.
- 2) Central Processing unit (CPU) - Once information is entered into computer by input devices,

processes process it. It control centre of computer. It fetch instruction from memory and then interpret them so as to know what we have done.

- CPU has 3 main components -
 - 1) ALU (Arithmetic logic unit)
 - 2) CU (Control Unit)
 - 3) Memory

ALU perform mathematical calculation and take logical decision. Arithmetic calculation include addition, subtraction, multiplication and division. Logical decision involves comparison of two data and see result which one is larger or smaller or equal.

Control unit co-ordinate and control dataflow in and out of CPU and also control all operations of ALU, memory, input output devices.

Memory - A register is a temporary unit of memory in CPU. These are use to store the data which is directly use by processor.

Memory attach to CPU is used for storage
⇒ of data and instructions and is called internal memory, also called primary memory

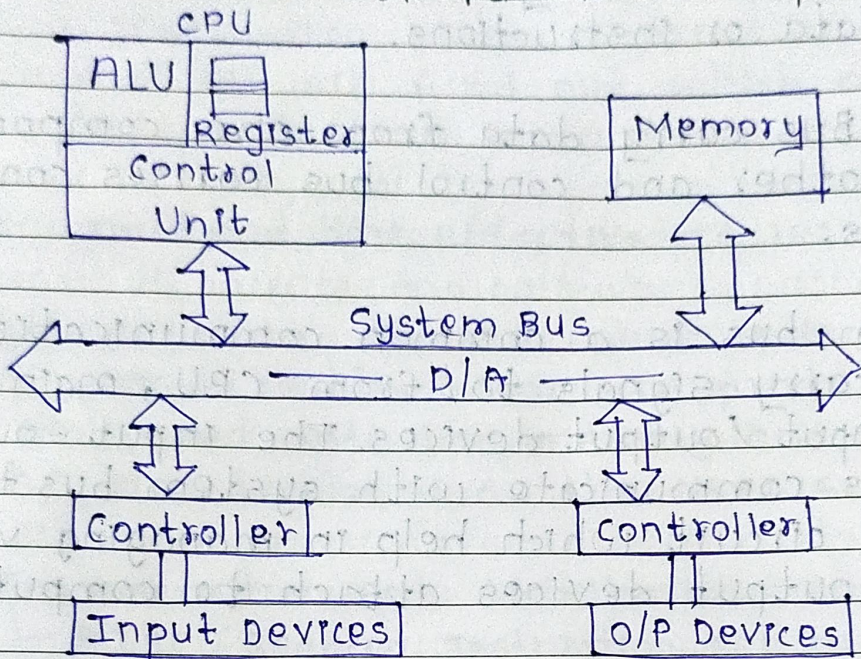
Ex. RAM.

Second one is secondary memory which store data permanently into computer.

3) Output unit consists of output devices that are attached with computer. It convert binary data coming from CPU to human understandable form. Examples are monitor, printer.

17/10/23

* Interconnection of functional component —



A computer consists of input unit that takes input, a CPU that process the input data and an output unit that produce output. All these devices communicate with each other through a common bus.

A bus is a transmission path which is made of a conducting wire over which data or information in form of electric signals is pass from one component to another in a computer.

The bus can be of 3 types -

1) Address Bus

2) Data Bus

3) Control Bus

- Address bus carry the address location of the data or instructions.
- Data Bus carry data from one component to another and control bus carries control signals.
- System bus is a common communication path that carry signals to/from CPU, main memory and input/output devices. The input-output devices communicate with system bus through control circuit which help in managing various input-output devices attach to computer.

* Instruction set Architecture - (ISA)

ISA is a part of abstract model of a computer that define how the CPU is controlled by the software. ISA act as an interface between hardware and software. It specify both what processor is capable of doing as well as how it gets done.

In a basic computer each instruction cycle consist of following phases

- 1) Fetch - The fetch operation is use for taking instruction those are given by user and instruction those are store into main memory will be fetch by using registers.
- 2) Decode - Decode operation is use for interpreting the instruction means instructions are decoded means CPU will find out which operation is to be perform on instruction.
- 3) Read - Read the effective address from memory.
- 4) Execute - The execute operation is perform by CPU and results those are produced by CPU are stored into the memory and after that they are displayed on the user screen.

19/10/22

- Instruction set Architecture
 - └ RISC (Reduced Instruction set cycle)
 - └ load
 - └ Evaluate
 - └ Store
 - └ CISC (Complex Instruction set cycle)
- Characteristics of RISC -

The main idea behind this is to make hardware simpler by using an instruction set composed of a few basic steps for loading, evaluating and storing operations just like a load command will load the data, a store command store data

- 1) simpler instruction, hence simple instruction loading
- 2) instruction comes under size of one word

- 3) instruction takes a single clock cycle to get executed
- 4) more general purpose registers are used
- 5) simple addressing modes
- 6) fewer data types used
- 7) a pipeline can be achieved

• CISC -

The main idea is that a single instruction will do or loading, evaluating and storing operation just like a single command will do loading, evaluating and storing it.

• Characteristics of CISC -

- 1) complex instruction, hence complex instruction decoding
- 2) instruction are larger than one word size
- 3) instructions may take more than a single clock cycle to get executed
- 4) less no. of general purpose register as operation get perform in memory itself
- 5) complex addressing modes
- 6) more data types

• CISC example -

There will be a single command or instruction which will perform a task

• RISC example -

Here programmer will first write load command to load data in register then it will use a suitable operator & store result in desire location

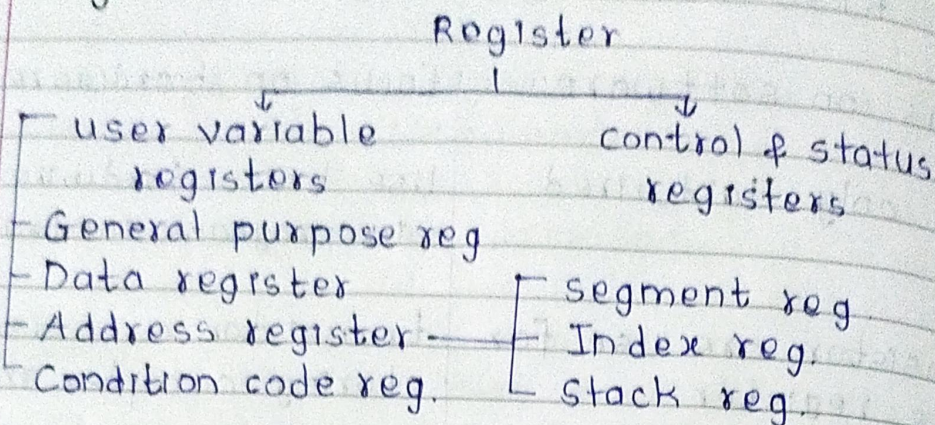
30/10/21

RISC

CISC

1) focus on software	focus on hardware
2) Use only Hardwired	Use both Hardwired and microprogram control unit
3) transistor are use for more register	transistor are use for storing complex instruction
4) fixed size instruction	complex ^{variable} size instruction
5) Can only perform register to register arithmetic operations	Can perform register to register, register to memory and memory to memory
6) require more no. of registers	require less no. of registers
7) code size is large	code size is small
8) an instruction executed in a single clock cycle	instruction takes more than one clock cycle
9) an instruction fit in a word	instructions are larger than size of one word

* Registers -



A register is a volatile memory system that provide the CPU with rapid access to information it is immediately using.

Registers are very useful components of CPU, an instruction works fastest when its operand are stored in CPU register. The register in CPU program perform 2 roles -

1) User Variable register -

These register can be use during assembly language programming.

2) Control & status register -

These register are use in control unit, controller can control operations of CPU to these registers. Certain registers are made for exclusive use of O.S.

Types of user variable registers -

① General purpose register -

It can be use for variety of data. Sometimes they are use for holding of operands of an instruction. Sometime general purpose register can be use for addressing purpose.

② Data register -

It may be use exclusively for holding data and they cannot be used for addressing. A register used in micro-computer to temporarily store data be transmitted to / from peripheral devices.

③ Address register -

Address register are use exclusively for the purpose of addressing data.

i) Segment register -

Segment register hold the address of the base of a segment. Many machines use segmentation for memory management. These segments are accessed using segment register.

ii) Index register -

It is use for indexing the addresses of the segments.

iii) Stack register -

Stack pointer is a dedicated register that point to the top of the stack. Stack pointer is auto-incremented or auto-decremented during push or pop operations.

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④ Condition code register -

It is manipulated by CPU to store the effect of execution of various arithmetic and logical operation.

Ex. An arithmetic operation may produce +ve, -ve, zero or overflow result then some fields of flags include the following

i) Zero bit -

contain : It is set when result is zero

ii) Sign bit -

If sign bit is set then result will be -ve

iii) Carry bit -

It is set if an operation resulted in carry out (addition) or borrowing (subtraction), out of a high order bit

iv) overflow bit -

It is used to indicate arithmetic overflow

v) Interrupt Enable / Disable -

It is used to enable / disable an interrupt.